TIDES

Harmonic tide prediction, offline, for 8,334 stations

Version 1.2 — no network, no subscription, nothing that expires. With six coastal work calculators that show every step.

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1. What this is, and why it is built this way

TIDES predicts the tide at a chosen station: the day's high and low water, the curve between them, and the two questions a boat actually asks of a tide. It runs as a POSIX shell tool and as a browser app, from the same arithmetic.

Four constraints shaped it, and each has a consequence worth knowing.

1. It must work with the antenna down. Consequence: the harmonics ship with the program. There is no service to call, no subscription to lapse, and no year in which it stops working.

2. The arithmetic must be checkable. Consequence: every calculator prints its working, line by line, and the browser prints the *same lines as the shell, character for character*. A test compares them on every worked case. One set of arithmetic to trust, checked two ways.

3. A prediction must never be dressed up as more than it is. Consequence: a subordinate station says **SECONDARY** on the screen and names its reference. An extrapolated figure says **EXTRAPOLATED**. A course that cannot be made good says so instead of printing a number.

4. It is a teaching tool as much as an answer. Consequence: the working is the point. A student who has watched the arithmetic once can do it with a pencil on a chart, which is the condition that matters when the battery is flat.

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2. Choosing a station, and why not a position

A tide is not computable from a position. It depends on the shape of the coast and the depth and resonance of the basin, so every place needs constants that somebody measured there. The nearest station in a straight line can be on the wrong side of a headland, and a river station can be nothing like the water outside it.

Two kinds of station, and the difference matters:

Reference	constants measured there; the curve is a real synthesis
Subordinate	no constants; carries time and height corrections to a reference station, exactly as a paper tide table has always worked

For a subordinate station the turns are the reference's turns corrected, and the curve between them is a half cosine. **There is no honest way to synthesise a continuous curve**

out of offsets, and this program does not pretend to — it says SECONDARY and names the reference.

Finding one

A tide cannot be computed from a position. It depends on the shape of the coast and the depth and resonance of the basin, so every place needs constants somebody measured there. You pick a *station*, not a place — and the nearest one in a straight line can be on the wrong side of a headland.

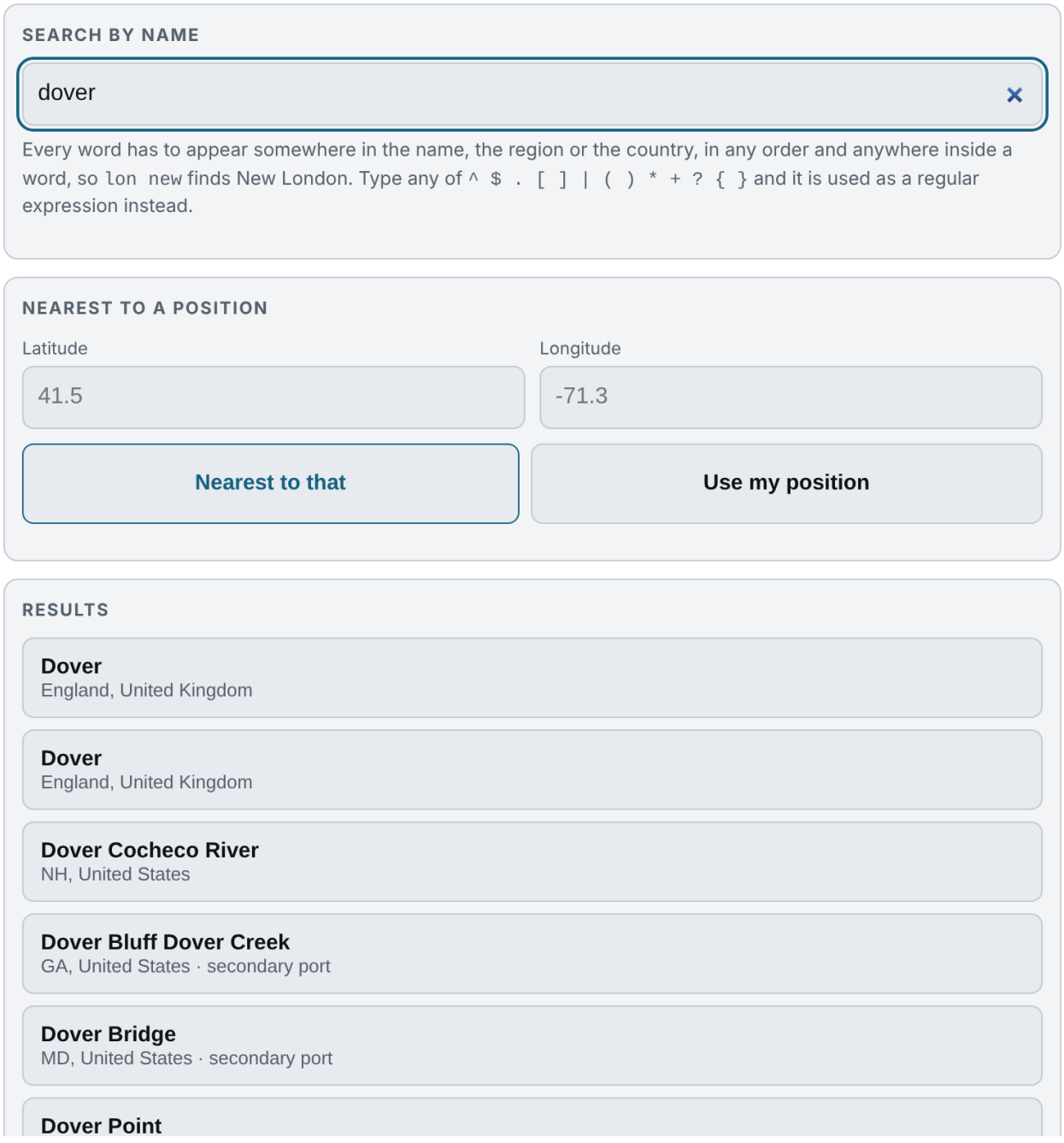


Figure 2 - the station picker. The reason a position is not enough is at the top of it, where it is read rather than looked up.

Nobody guesses a station's exact name: the database calls a place `NEW LONDON State Pier`. So type part of it and pick from the list. Every word has to appear somewhere in the name, state or country, in any order and anywhere inside a word, so `lon new` finds New London as well as `new london`.

The text becomes a regular expression the moment it contains any of `^ $ . [ ] | ( ) * + ? { } \` —

```
tides find "^st mary"        names that start with St Mary
tides find "bay$"            names that end in Bay
tides find "falmouth|mystic" either one
```

A pattern that is not valid is searched as plain text rather than refused.

Times are the station’s own standard time. No summer time, exactly like a printed table. GMT for the UK, all year.

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3. The two questions a boat asks



Figure 1 - a day at a reference station: the turns, the curve, and the heights above chart datum. The banner says REFERENCE, so these are computed from constants measured here rather than offset from somewhere else.

Is there enough water?

Charted depth **plus** the tide, against your draught and whatever you want under the keel — with the stretches of the day when it is there.

Does the mast go under?

Charted clearance **minus** the tide, against your air draught.

The signs are opposite, and that is the trap. Charted depths are measured *below* datum, so the tide adds to them. Charted vertical clearances are measured *above* HAT, so the tide takes them away. Getting that pair the wrong way round is the classic way to put a mast into a bridge. They are therefore separate functions in the engine and separately labelled on the screen, rather than one function with a sign argument.

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4. Command reference (shell)

`tides`	the menu
`tides near <lat> <lon>`	the stations nearest a position
`tides find <text>`	search by name
`tides use <id>`	choose a station
`tides today [date]`	the day’s table, curve and depth helper
`tides sky [date]`	the same, with the moon and sun panel
`tides work <name> k=v …`	coastal navigation, working shown
`tides where`	which station is chosen, and where the files are
`tides day|night|plain`	colour mode
`tides about`	what this is, where the data comes from
`tides version`	

Dates are `YYYY-MM-DD`.

In the browser the same things are on screen, and a URL opens any calculator directly — see section 6.

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5. The six coastal calculators

Each shows its working step by step, because the point is teaching. They are on the day screen and the station picker in the browser, and are `tides work <name>` in the shell.

5.1 `twelfths` — height at a time

The hourly twelfths table (1, 2, 3, 3, 2, 1), the height by the rule of twelfths, and the height by the cosine method

```
    h = h1 + (h2 - h1) x (cos A + 1) / 2
    A = 180 x ((t - t1) / (t2 - t1) + 1) degrees
```

with the difference between them. Two forms: give the four turn values, or give a station and let the harmonic prediction supply the turns — then you get three answers side by side, twelfths, cosine and the harmonic height itself.

HW before LW is a **falling** tide. The pair of turns used is the one less than twelve hours apart, so a tide crossing midnight works.

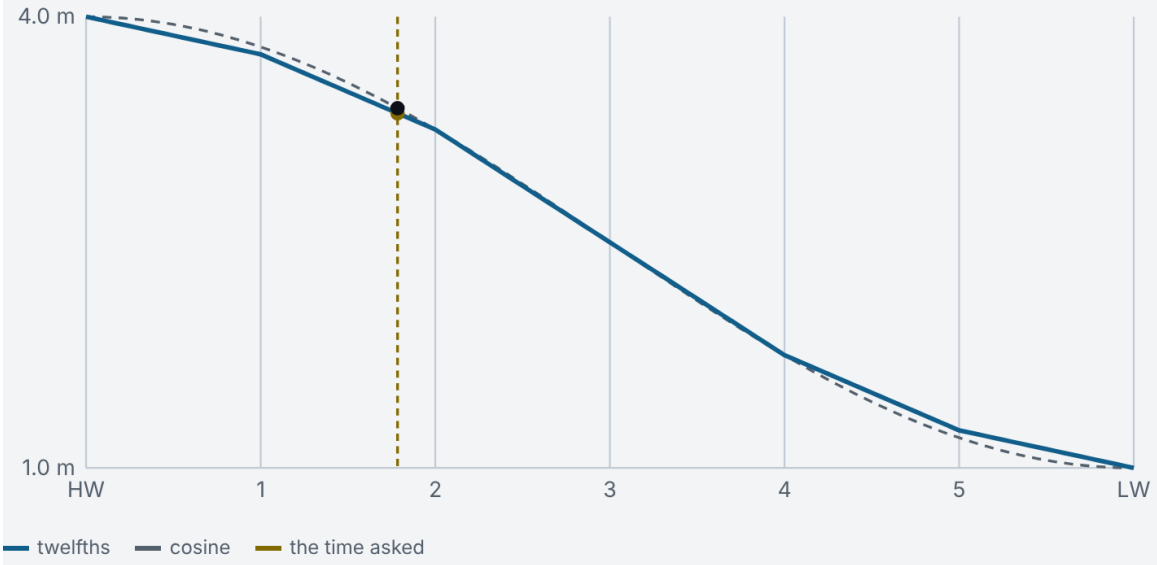
```
RULE OF TWELFTHS, AND THE COSINE CURVE
------------------------------------------------------------------------
HW 23:10  4.00 m   to   LW 05:20  1.00 m      a falling tide
range 3.00 m over 6 h 10 min, so a tidal hour here is 1 h 02 min
height wanted at 01:00: 1 h 50 min after HW = 1.78 tidal hours

1  The twelfths, tidal hour by tidal hour
   hour  time     this hour        so far    height
     0   23:10        -           0/12     4.00 m
     1   00:12    1/12  -0.25 m    1/12     3.75 m
     2   01:13    2/12  -0.50 m    3/12     3.25 m
     3   02:15    3/12  -0.75 m    6/12     2.50 m
     4   03:17    3/12  -0.75 m    9/12     1.75 m
     5   04:18    2/12  -0.50 m   11/12     1.25 m
     6   05:20    1/12  -0.25 m   12/12     1.00 m

2  At 01:00: 1 tidal hours give 1/12, and 0.78 of hour 2's 2/12 adds 1.57/12
   2.57/12 of 3.00 m = 0.64 m, so h = 4.00 - 0.64 = 3.36 m

3  The cosine curve:  h = h1 + (h2 - h1) x (cos A + 1)/2
   where A = 180 x (fraction of the time from turn to turn + 1) degrees
   fraction of the way from HW to LW = 1 h 50 min / 6 h 10 min = 0.297
   A = 233.5 deg, cos A = -0.595
   h = 4.00 + (-3.00) x (-0.595 + 1)/2 = 3.39 m

4  Twelfths minus cosine: -0.03 m.  The twelfths are a staircase laid over
   the cosine: they agree at the turns, at half tide and at hours 2 and 4,
   and differ most an hour either side of a turn.
```




```
ANSWER   twelfths 3.36 m    cosine 3.39 m
------------------------------------------------------------------------
```

Figure 3 - the rule of twelfths worked out, with the cosine height beside it and the difference between the two stated. The working is the deliverable: a reader who has watched it once can do it on paper with the tide table alone.

```
tides work twelfths lw=05:12 lwh=0.8 hw=11:24 hwh=4.2 t=08:30
```

5.2 `secondary` — a secondary port from its standard port

Give the HW part, the LW part, or both.

1. **Time differences** are interpolated round the clock. The almanac gives them for two standard-port times and the same again twelve hours later, so times are taken **modulo 12 hours** and interpolated linearly round that circle.
2. **Height differences** are interpolated by the **standard port's HW height between MHWN and MHWS** — not by range. That is the RYA method.
3. Outside those limits the line is carried on and the working says **EXTRAPOLATED**.
4. The time difference is rounded to the nearest minute.

5.3 `stream` — a rate between neaps and springs

```
    rate = nps + (range - nprange) / (sprange - nprange) x (spr - nps)
```

Beyond springs or below neaps the line is extrapolated and flagged. A rate that would come out below zero is taken as 0.0 kn and says so.

```
tides work stream spr=2.6 nps=1.4 range=4.1 sprange=5.9 nprange=3.1
```

5.4 `atlas` — which page of the tidal stream atlas

Which page applies, as hours from the **nearest HW** at the atlas's standard port: `HW -3` … `HW` … `HW +6`. Each page covers thirty minutes either side of its hour, and **exactly half past goes to the later page** — HW +0:30 is page HW +1, HW −0:30 is page HW.

Then the range of the tide you are in, and where it sits between mean neaps and mean springs. The means are the Admiralty definition taken from the station's own constants, springs 2(M2 + S2) and neaps 2(M2 − S2) — given only where the tide is semidiurnal, which is a form factor (K1 + O1)/(M2 + S2) up to 0.25. At a secondary port, or a mixed or diurnal tide, the working says the means are not available **and why**, and gives the range alone.

5.5 `cts` — course to steer

Over several hours the stream vectors are **summed and the triangle solved once**, as taught — not solved hour by hour.

1. The working gives the water track and heading to a tenth of a degree; the answer is the whole degree to steer.
2. **Leeway is applied into the wind.** `wind=port` means the wind is on the port side, so she is set to starboard and heading = water track − leeway. `wind=stbd` is the other way.
3. Then magnetic = true − variation, and compass = magnetic − deviation. Variation and deviation are **east positive**, so `var=-2` is 2° W.
4. If the stream sets her across the track faster than she sails, the answer is **"cannot make good this track"**, with the working that shows why.

   tides work cts track=045 stw=6 streams=135:1.5,150:2.0 leeway=5 wind=port

```
COURSE TO STEER
-------------------------------------------------------------------
track to make good 045 T, 6 kn through the water, 2 hours of stream

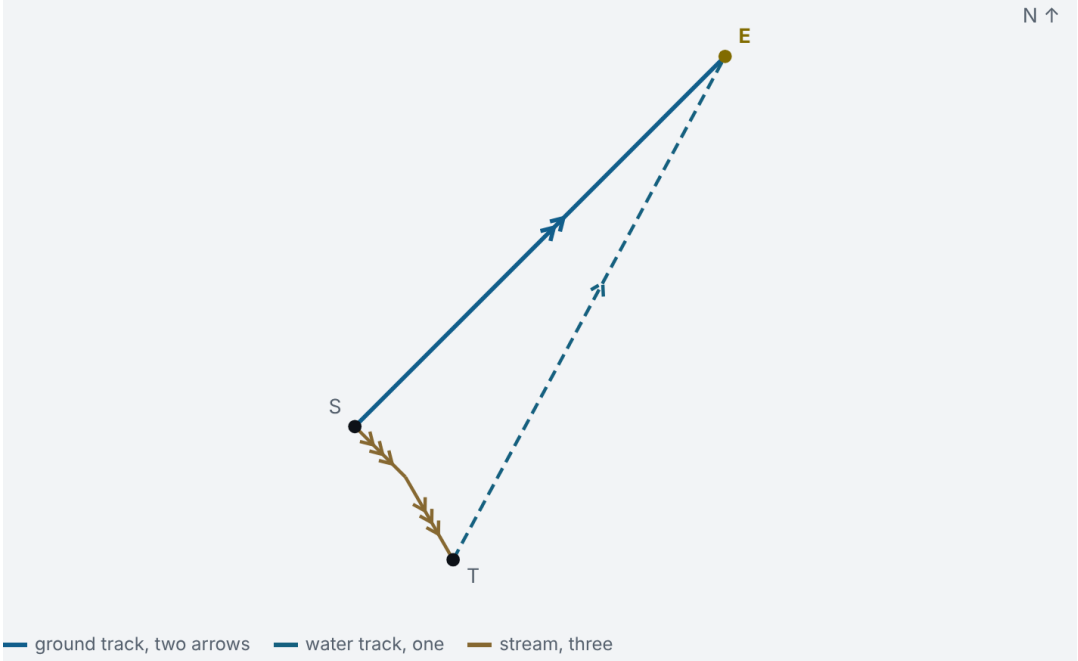
1  The stream, hour by hour, added up
   hour 1   set 135 T  1.5 kn   -> 1.50 nm
   hour 2   set 150 T  2.0 kn   -> 2.00 nm
   together the water moves 3.47 nm toward 144 T in 2 h

2  She sails 6 kn x 2 h = 12.00 nm through the water

3  Across the track the stream sets her 3.43 nm to starboard;
   along it, 0.52 nm against her
   to cancel it, point to port: sin x = 3.43 / 12.00 = 0.286, x = 16.6 deg
   water track = 045 - 16.6 = 028.4 T

4  Made good along the track: 12.00 x cos 16.6 + (-0.52) = 10.98 nm in 2 h
   speed over the ground 5.49 kn

5  Leeway 5 deg, wind on the port side: she slides to starboard,
   so steer 5 deg to port of the water track: 028.4 - 5 = 023.4 T
   steer the whole degree: 023 T
6  Variation 2 W: 023 T + 2 = 025 M
   deviation 1 E: 025 M - 1 = 024 C
7  12 nm at 5.49 kn takes 2 h 11 min
   longer than the 2 h the streams cover: this assumes the same
   stream carries on
```




```
ANSWER   steer 023 T  025 M  024 C    SOG 5.5 kn    2 h 11 min for 12 nm
-------------------------------------------------------------------
```

Figure 4 - that example worked: the streams summed, the triangle solved once, leeway applied into the wind, then variation and deviation. The drawing is the same triangle, with the ground track, the water track and the stream each carrying their own number of arrows so they are told apart with no colour at all.

5.6 `setdrift` — set and drift from a DR and a fix

The DR runs from the start along the water track; the vector from the DR to the fix over the time run is the set and drift, and rate = drift / hours. Plane sailing at the mean latitude, which is right for coastal distances — it is **refused beyond latitude 80**. The antimeridian is handled.

With no leeway given, the set and drift found **includes any leeway too**, and the working says so.

6. Linking into a calculator, and the parameter contract

In the browser, a URL opens any calculator directly:

```
?work=cts&track=045&stw=6&streams=135:1.5,150:2.0&lesson=l3-2025-4
```

The coastal navigation course deep-links into these from its lessons, so **the parameter names are a contract**: a name is never renamed or given a new meaning. `docs/web/tides/WORK.md` is the authoritative list, with every unit and default.

Common rules

angles	degrees **true**, 0–360 (`360` reads as `0`)
speeds	knots
heights, ranges	metres
distances	nautical miles
times	`HH:MM`, or `HHMM`
positions	`lat,lon` decimal degrees, **south and west negative**
dates	`YYYY-MM-DD`
variation, deviation	degrees, **east positive**

A `+` in a query string is a space. A browser reads `tdhw2=+5` as `tdhw2= 5`. Write positive values unsigned, or as `%2B5`.

Every value is checked. An unknown parameter, a value that is not a number, an angle outside 0–360, an impossible date: the page shows one plain sentence naming the parameter and works nothing. **It never shows a partial answer from bad input.**

7. Where the numbers come from

1. **Seventy constituents**, with their satellite and shallow-water terms, UTide-generated. They are **dumped from the engine's own runtime strings** and never retyped by hand.
2. **The browser is a port of the shell engine,** line by line, and checked against it. Heights agree to about a femtometre and the turning times are exact.
3. **The shell engine is held to an independent Python working** of each method. So the browser is checked against the awk, and the awk against something that shares none of its code.
4. **The pictures are the one place the builds differ on purpose:** the shell draws characters, the page draws SVG from the same numbers, and the SVG's points are read back and checked against the geometry they claim.

8. Offline, and how the station data is shipped

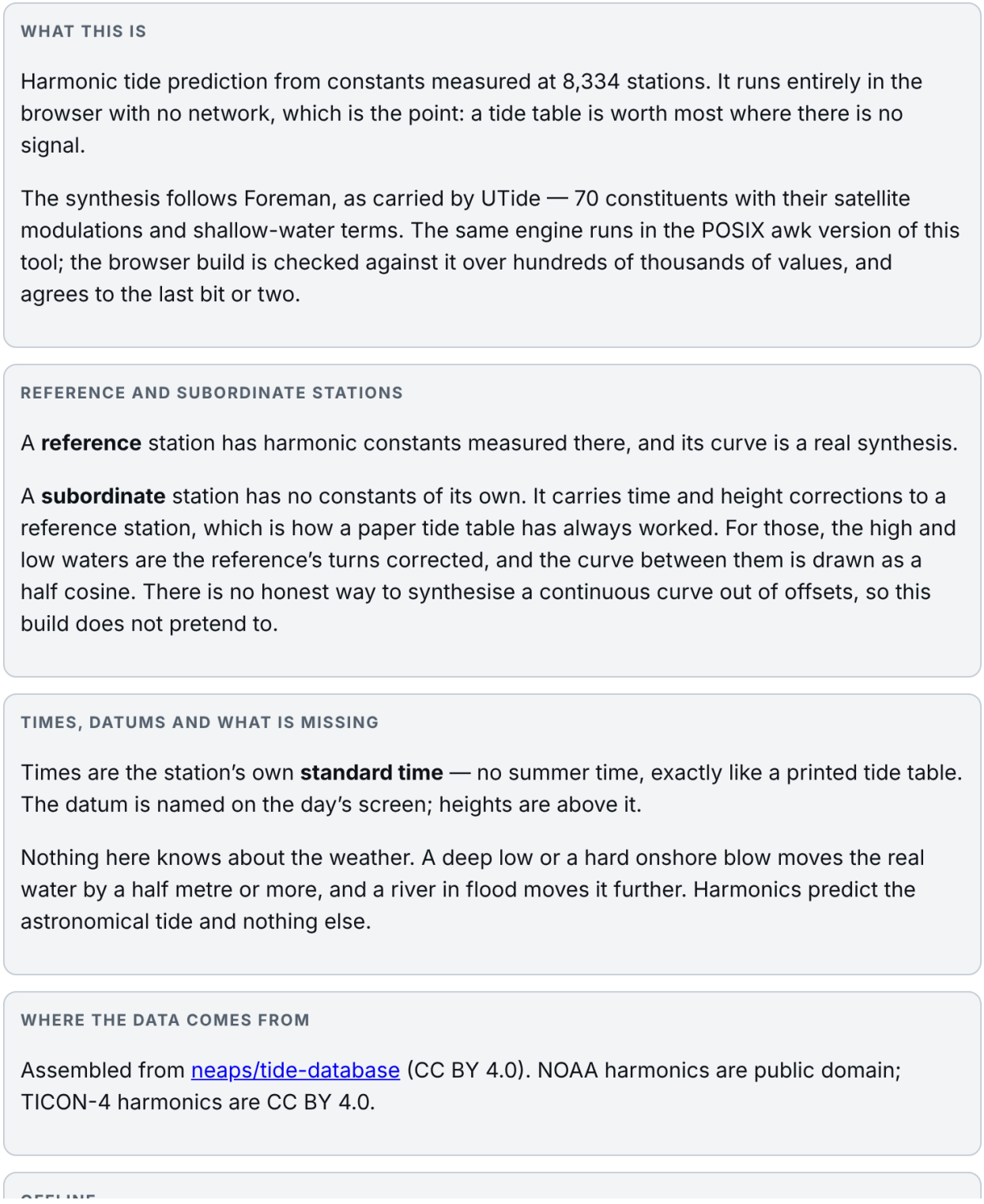


Figure 5 - the About screen, which says what is actually in the cache rather than what should be. On a first visit it says so plainly: not installed, served from the network.

The station file is 2.9 MB, which is not something to download before the app will draw anything. So an index of every station — name, region, country, position, and the **byte offset and length** of its line — loads at startup, and one station's constants come out of the big file with a range request of a few hundred bytes.

Three things about that are easy to get wrong, and are worth knowing if you ever see odd behaviour:

1. **The offsets are byte offsets.** Four station names are not ASCII, so the file is held and sliced as bytes. Slicing a decoded string by a byte offset lands mid-line for every station after Saint Barthélemy.
2. **A partial response must never be cached as the whole file.** Stored under the main file's name it would make every later read return a few hundred bytes, and the failure looks like corrupt data rather than a caching mistake.
3. **The station file is cached on first successful fetch,** not at install — one large file in the install list fails the whole install on a flaky connection, and then nothing at all is cached. The About screen has a button that fetches it with a progress bar.

Prove it once at the dock: open a station, turn the network off, relaunch from the icon.

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9. Accuracy, and what it does not know

Nothing here knows about the weather. A deep low or a hard onshore blow moves the real water by half a metre or more, and a river in flood moves it further. Harmonics predict the **astronomical** tide and nothing else.

A few other limits worth carrying in your head:

1. **A subordinate station's curve is a half cosine** between corrected turns. It is a tide table's answer, not a synthesis.
2. **Mean springs and neaps are only given for a semidiurnal tide** at a reference station. Elsewhere the program says why it will not give them.
3. **Set and drift is plane sailing** at the mean latitude, refused beyond 80°.
4. **The twelfths rule is a rule of thumb**, which is why the cosine method and — at a station — the harmonic height are printed beside it. Where they disagree, the harmonic height is the better number.

The display: day, dark and night

The browser build has three display modes. They are not decoration: one of them exists to protect your night vision, and choosing the wrong one at the wrong moment costs you something real.

Mode	What it is for
NIGHT	Red on black, for a dark-adapted eye. The default.
DARK	Near-black with light text, for below decks, dusk, or indoors.
DAY	White with near-black text, for daylight and for printing.

Why night is the default

A tool you open on deck in the dark must not start bright. A bright panel costs a dark-adapted eye roughly **twenty minutes** to recover from, and twenty minutes is a long time to be unable to see a mark or a ship's lights properly. So the first time you open any of these tools you get night, and from then on you get whatever you last chose.

Three buttons, not a cycle

The three modes are three separate buttons, and the one you are in is shown as selected. That is deliberate.

1. **A cycle can take you somewhere you did not intend.** One stray press with a cycling control can land you on DAY at three in the morning.
2. **A cycle cannot tell you where you are**, only what the next press would do. Three buttons show the current mode at a glance without changing it.
3. **Going darker is always one tap.** DARK and NIGHT are immediate from anywhere, because going darker never costs you anything.
4. **Going from NIGHT to DAY takes a deliberate half-second press.** Hold the DAY button and it fills as it goes; let go early and nothing happens. Nothing bright appears on the screen until the change actually completes. That one transition is guarded because that one transition is the only one that costs you your dark adaptation.

The setting follows you

The mode is remembered for all of the tools together, and for the course pages that embed them. Set night in one and you are in night in the next. It is remembered in the browser, so it survives closing the app.

It is right before the page draws

The mode is set in the first instructions the page runs, before any of it is painted. This matters on a first visit over a poor connection: set it later and the page would appear in one palette and then jump to another, which at night is the difference between keeping your night vision and losing twenty minutes of it. If nothing has been stored yet, or whatever was stored cannot be read, the page opens in **night**.

The controls follow the palette too

Sliders, scroll bars and the other controls the browser draws are told which world they are in, so they are drawn in the mode you chose rather than the browser's own light default. Without that, a night screen can carry a bright white slider across the middle of it.

If you use High Contrast Mode

On Windows, High Contrast replaces every colour on the page with the ones your system theme specifies. That applies here too: the tools do not override it, because it is your setting and you chose it.

Nothing is lost by it. Where a colour carries meaning — the navigation lamps in colregs — the letter beside it says the same thing, so a lamp is still identifiable when the hue has gone.

One thing to know if you need both. High Contrast wins over night mode. If you need night vision on deck *and* high contrast to read, choose a dark high-contrast theme in Windows; the tools will follow it.

What the display can and cannot do in sunlight

Worth knowing before you blame the app on a bright day.

In sunlight, light reflecting off the glass lands on the dark parts of the screen and the light parts alike. That lifts everything together and flattens the difference between them, so the contrast you actually see collapses no matter what colours the page uses:

Condition	Body text, as seen
Indoors	about 10:1
Overcast on deck	about 6:1
Bright	about 3.5:1
Full sun on the glass	about 2.3:1

At the extreme, every colour scheme converges. Pure black on pure white measures the same 2.3:1 in full sun as the day mode does, because the glare swamps the difference. A different palette is not the answer, and nor is a different shade of white: across every plausible choice of background the contrast changes by under 4%, which is below what an eye can see.

What does work, in order:

1. **Turn the screen brightness up.** This is the one large lever, and it is the device’s, not the app’s.
2. **Shade the screen** with a hand, your body or the sprayhood. Removing the reflection is worth more than any change the app can make.
3. **Use DAY mode**, which at least puts the maximum luminance the panel can give behind the darkest text it can draw.

A note for printing

Documents and their figures are made in **day** mode. A screen captured in dark or night mode prints as a solid block of toner and loses the fine type in a working, so the printed pages show the day display even if that is not the one you use at sea.

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10. Cautions

This is a training and cross-checking aid, not a substitute for judgement.

Where this program and the published tide tables for the port differ, **the published tables are right and this is wrong.** The same goes for the almanac, the tidal stream atlas and the chart.